

# Problems with Neural Networks

Vanishing & Exploding gradients

- weight init
- Activation function
- Batch Normalization
- Gradient Clipping

Not enough data

- transfer learning
- Unsupervised pre-training

Slow training

- Optimizers (Adam)
- Learning rate scheduler

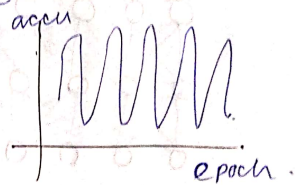
Overfitting

- $L_1, L_2$  Regularization
- Dropouts

While using unscaled data in ANNs →

there most probably the algorithm cannot converge to the best solution.

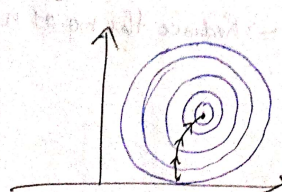
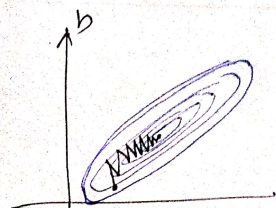
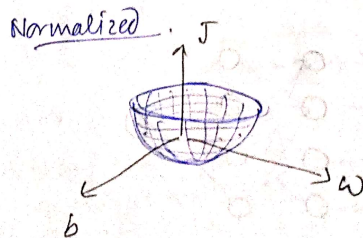
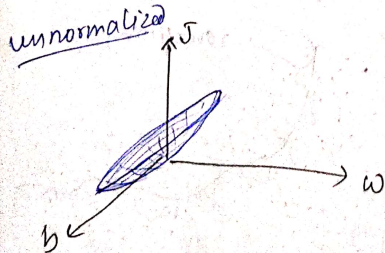
you can see an oscillatory motion in the accuracy curve



age range, range, range  
 $x_1, x_2$  (Salary)

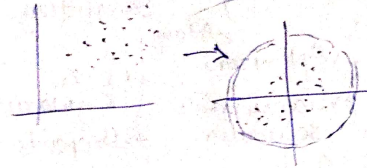
$$w_1, w_2 \quad w_2 = w_1 - \eta \frac{\partial L}{\partial w_1}$$

focus on this weight more because  $x_2$  is more focused.



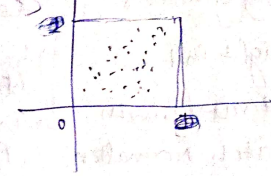
Standardize

$$\frac{x_i - \mu}{\sigma}$$

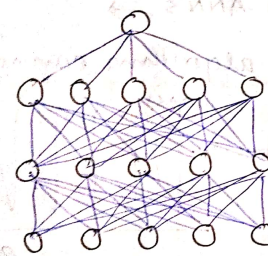


Normalize

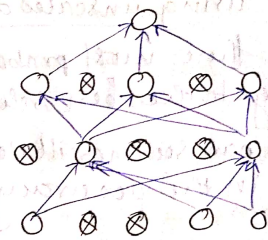
$$\frac{x_i - x_{min}}{x_{max} - x_{min}}$$



## The concept of dropouts



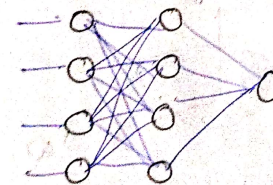
Standard NN



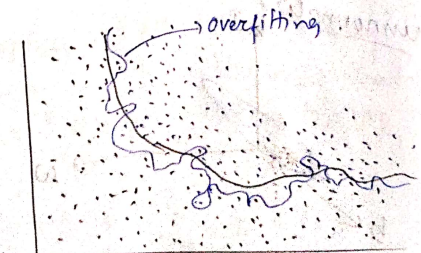
Applying dropout

randomly drop some neurons in each epoch.

Actually training a fixed data but training with different NNs in each epoch.



→ Reduce the no. of nodes



overfitting